

Brandon Forys

PhD, Psychology (Cognitive Science), University of British Columbia

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Education and training

University of British Columbia, Vancouver, BC, Canada, PhD in Psychology 2022-2026

Supervisors: Dr. Rebecca Todd & Dr. Alan Kingstone

PhD dissertation: *What predicts cognitive effort? Optimality, preferences, and strategies.*

University of British Columbia, Vancouver, BC, Canada, M.A. in Psychology 2020-2022

Supervisors: Dr. Rebecca Todd & Dr. Alan Kingstone

Master's thesis: *Gender moderates the relationship between mood disorder symptoms and effortful avoidance performance.*

University of British Columbia, Vancouver, BC, Canada, B.A. Honours in Psychology 2016-2020

Supervisors: Dr. Rebecca Todd (Honours), Dr. Kiran Soma (Honours), & Dr. Timothy Murphy

Honours theses: *Modulating effort cost in human avoidance behaviours.; A novel steroidogenic model for reward-seeking behaviour.*

Honours and awards

Canada Postdoctoral Research Award, NSERC 2027-29

Faculty of Arts Graduate Award, UBC 2025

Faculty of Arts Graduate Award, UBC 2023

Stanley Coren Prize in Psychology, UBC 2023

Four Year Fellowships (4YF) For PhD Students, UBC 2022-2026

President's Academic Excellence Initiative PhD Award, UBC 2022-2026

Canada Graduate Scholarship – Doctoral, NSERC 2022-2025

Faculty of Arts Graduate Award (deferred from 2020), UBC 2021

Canada Graduate Scholarship – Master's, NSERC 2020

NSERC Undergraduate Student Research Award, NSERC/UBC 2020

Faculty of Arts Graduate Award, UBC 2020

Trek Excellence Scholarship for Continuing Students, UBC 2019

ARTA Scholarship, Alberta Retired Teachers' Association 2019

Wesbrook Scholar, UBC 2019

HSBC Emerging Leader Scholarship, UBC 2019

Suedfeld Scholar Award, UBC Psi Chi 2019

PSYC 217 Poster Award, 1st Place, UBC Psychology 2018

Student Scholarship in Arts, UBC Faculty of Arts 2018

Trek Excellence Scholarship for Continuing Students, UBC 2017

Stephen Straker Arts One Prize, UBC Arts One Program 2017

University of BC Sopron Memorial Scholarship, UBC 2016

Dean's List, UBC Faculty of Arts 2016-17, 2017-18, 2018-19, 2019-20

Publications and presentations

JOURNAL ARTICLES AND PREPRINTS

1. Hamden, J. E., Ainsworth-Cruikshank, G., Gray, K. M., Ye, J., Kachkovski, G. V., **Forys, B. J.**, Sie, C., Silva, J. A. F. D., Salehzadeh, M., Ghosh, S., Harder, K. W., & Soma, K. K. (2026). The parental intestinal microbiome modulates systemic and local glucocorticoid levels in neonatal mice. *PLOS ONE*, 21(9), e0357469. <https://doi.org/10.1371/journal.pone.0357469>
2. Tong, L. C., **Forys, B. J.**, Hales, C. A., Clark, L., & Winstanley, C. A. (2026). Anterior insula activity increased by cued risky wins in healthy volunteers (p. 2026.06.30.735658). *bioRxiv* (preprint). <https://doi.org/10.64898/2026.06.30.735658>
3. **Forys, B. J.**, Winstanley, C. A., & Todd, R. M. (2025). Short-term memory capacity and chronic stress levels predict cognitive effort choice as a function of reward level and effort demand (p. 2025.07.24.666659). *bioRxiv* (preprint). <https://doi.org/10.1101/2025.07.24.666659>
4. Tomm, R. J., **Forys, B. J.**, Kalenteridis, L., Daly, I., Terpstra, A. R., Clark, L., Chakrabarty, T., Floresco, S. B., & Todd, R. M. (2025). Depression levels are associated with reduced capacity to learn to actively avoid aversive events in young adults. *eNeuro*, 12(9). <https://doi.org/10.1523/ENEURO.0034-25.2025>
5. **Forys, B. J.**, Qi, E.**., Todd, R. M., & Kingstone, A. (2025). Hear it here: Built environments predict ratings and descriptions of ambiguous sounds. *PLOS ONE*, 20(1), e0316187. <https://doi.org/10.1371/journal.pone.0316187>
6. Ehlers, M. R., Kryklywy, J. H., Beukers, A. O., Moore, S. R., **Forys, B. J.**, Anderson, A. K., & Todd, R. M. (2024). Valenced tactile information is evoked by neutral visual cues following emotional learning. *Imaging Neuroscience*, 2, 1–16. https://doi.org/10.1162/imag_a_00320
7. **Forys, B. J.**, Winstanley, C. A., Kingstone, A., & Todd, R.M. (2024). Short-term memory capacity predicts willingness to expend cognitive effort for reward. *eNeuro*, 11(7). <https://doi.org/10.1523/ENEURO.0068-24.2024>
8. Kryklywy, J. H., **Forys, B. J.**, Vieira, J. B., Quinlan, D. J., Culham, J. C., & Mitchell, D. G. V. (2023). Dissociating representation of affect and motion in visual cortices. *Cognitive, Affective, & Behavioural Neuroscience*, 1-24. <https://doi.org/10.3758/s13415-023-01115-2>
9. **Forys, B. J.**, Tomm, R. J., Stamboliyska, D.**., Terpstra, A. R., Clark, L., Chakrabarty, T., Floresco, S. B., & Todd, R. M. (2023). Gender impacts the relationship between mood disorder symptoms and effortful avoidance performance. *eNeuro*, 10(2). <https://doi.org/10.1523/ENEURO.0239-22.2023>
10. Xiao, D., **Forys, B. J.**, Vanni, M. P., & Murphy, T. H. (2021). MesoNet: automated scaling and segmentation of mouse mesoscale cortical maps using machine learning. *Nature Communications*, 12(1), 5992. <https://doi.org/10.1038/s41467-021-26255-2>
11. Hamden, J.E., Salehzadeh, M., Gray, K. M., **Forys, B. J.**, & Soma, K. K. (2021). Isoflurane stress induces glucocorticoid production in mouse lymphoid organs. *Journal of Endocrinology*, 251(2), 1-13. <https://doi.org/10.1530/JOE-21-0154>
12. Hamden, J. E.*, Gray, K. M.*, Salehzadeh, M., Kachkovski, G. V., **Forys, B. J.**, Ma, C., ... Soma, K. K. (2021). Steroid profiling of glucocorticoids in microdissected mouse brain across

development. *Developmental Neurobiology*, 81(2), 189–206. <https://doi.org/10.1002/dneu.22808>

13. **Forys, B. J.***, Xiao, D.*, Gupta, P., & Murphy, T. H. (2020). Real-time selective markerless tracking of forepaws of head-fixed mice using deep neural networks. *eNeuro*, 7(3). <https://doi.org/10.1523/ENEURO.0096-20.2020>

MANUSCRIPTS IN PREPARATION

1. **Forys, B. J.**, Winstanley, C. A., & Todd, R.M. (in preparation). Short-term memory capacity and chronic stress levels predict cognitive effort choice as a function of reward level and effort demand.
2. Tong, L. C., **Forys, B. J.**, Hales, C. A., Clark, L., & Winstanley, C.A. (under review). Anterior insula activity increased by large cued risky wins in healthy volunteers.

*Co-first authors

CONFERENCE PROCEEDINGS

1. **Forys, B. J.**, Moise, N.**, Todd, R. M., & Winstanley, C. A. (2026). Similar factors predict cognitive effort preferences when working for rewards vs. Avoiding losses. JoCN Forum. Presented at Cognitive Neuroscience Society Annual Meeting, 2026, Vancouver, BC. <https://doi.org/10.21428/8e6ba8ef.3e04c045>

POSTER PRESENTATIONS

1. Handwerker, D. A., Bandettini, P. A., Abbott, D. F., Bruno, K. L., **Forys, B. J.**, Garcia, M., Goodale, S., Markiewicz, C., Moia, S., Pinsard, B., Reddy, N. A., Rubin, M., Salo, T., Smith, M., Smith, R., Tahayori, B., & Uruñuela, E. (2026). Growing Tedana: a grassroots-leveraged platform for processing multi-echo fMRI data. Presented at Organization for Human Brain Mapping (OHBM) 2026, Bordeaux, France.
2. **Forys, B. J.**, Singh, A.**, Qi, E.**, Todd, R. M., & Kingstone, A. (2025). Ratings and interpretations of everyday sounds differ with room lighting levels. Presented at Psychonomic Society Annual Meeting, 2025, Denver, CO.
3. **Forys, B. J.**, Winstanley, C. A., Kingstone, A., & Todd, R. M. (2024). What predicts the choice to expend effort for reward? A translational study. Presented at Neuroscience 2024, Chicago, IL.
4. **Forys, B. J.**, Tomm, R. J., Kryklywy, J. H., Clark, L., Chakrabarty, T., Floresco, S. B., & Todd, R. M. (2023). Using a novel form of pattern component modeling with representational similarity analysis to model information learned in active and inhibitory avoidance and reward-seeking. Presented at International Conference on Learning and Memory, 2023, Huntington Beach, CA.
5. **Forys, B. J.**, Kryklywy, J. H., Ehlers, M. R., Moore, S. R., Beukers, A. O., Anderson, A. K., & Todd, R. M. (2022). Modelling representations of continuously shifting stimuli: a novel sliding window approach. Presented at Cognitive Neuroscience Society Annual Meeting, 2022, San Francisco, CA.

6. Zheng, J.**, **Forys, B. J.**, & Todd, R. M. (2022). Maladaptive Transfer of Cognitive Effort in Emotional Situations. Presented at UBC Psychology Undergraduate Research Conference, 2022.
7. **Forys, B. J.**, Tomm, R. J., Stamboliyska, D.**, Terpstra, A. R., Clark, L., Chakrabarty, T., Floresco, S. B., & Todd, R. M. (2021). Gender selectively moderates relationships between mood disorder symptoms and performance of effort related avoidance and reward-seeking tasks. Presented at Neuroscience 2021.
8. Tomm, R. J., **Forys, B. J.**, Daly, I. D., Terpstra, A. R., Clark, L., Chakrabarty, T., Floresco, S. B., & Todd, R. M. (2021). Depressive and anxiety symptom clusters in humans are related to distinct avoidance profiles in a cross-species translation of a go/no-go avoidance task. Presented at Neuroscience 2021.
9. Daly, I. D., Tomm, R. J., **Forys, B. J.**, Terpstra, A. R., Clark, L., Chakrabarty, T., Floresco, S. B., & Todd, R. M. (2021). Levels of depressive and anxiety symptoms in humans predict distinct patterns of reward-seeking: A cross-species translation of an appetitive go/no-go task. Presented at Neuroscience 2021.
10. Tomm, R. J., Daly, I. D., **Forys, B. J.**, Floresco, S. B., & Todd, R. M. (2021). Anxiety and depressive symptoms in humans are related to distinct avoidance profiles in a cross-species translation of a go/no-go avoidance task. Presented at the International Behavioural Neuroscience Society Meeting, 2021.
11. Sidarth, A.**, **Forys, B. J.**, & Todd, R. M. (2021). Cognitive control on reward-seeking behavior. Presented at UBC Multidisciplinary Undergraduate Research Conference, 2021, and UBC Parkinson's and Alzheimer's Targeted Hope and Support Conference, 2021.
12. Xiao, D., Gupta, P., **Forys, B.**, & Murphy, T. H. (2020). Real-time forepaw movement coded auditory feedback promotes motor skill learning in mice. Presented at UBC Psychiatry Virtual Research Day, 2020.
13. **Forys, B. J.**, Xiao, D., Gupta, P., Boyd, J. D., & Murphy, T. H. (2019). Short latency (~100 ms) markerless video tracking of body parts in mice using deep neural networks. Presented at Neuroscience 2019, Chicago, IL, and UBC Neuroscience Undergraduate Research Conference, 2020.
14. Xiao, D., **Forys, B. J.**, Tandun, R., & Murphy, T. H. (2019). Automated alignment and segmentation of mouse mesoscale brain images using machine learning. Presented at Neuroscience 2019, Chicago, IL, UBC Future of Health Research Day, 2019, and UBC School of Biomedical Engineering Symposium, 2019.
15. **Forys, B.**, Tobiansky, D. J., & Soma, K. K. (2019). A novel steroidogenic model for reward-seeking behaviour. Presented at UBC Psychology Undergraduate Research Conference, 2019.
16. **Forys, B.**, Xiao, D., Gupta, P., Boyd, J. D., & Murphy, T. H. (2018). Real-time markerless video tracking of bodyparts in mice using deep neural networks. Presented at UBC Brain Circuits Cluster, 2018, Neuroextravaganza, 2018, UBC Undergraduate Neuroscience Conference, 2019, and UBC School of Biomedical Engineering Symposium, 2019.
17. Tobiansky, D. J., Kachkovski, G., Enos, R. T., Schmidt, K. L., Ma. C., **Forys, B.**, Hamden, J. E., Jalabert, C., Floresco, S. B., Murphy, E. A., Soma, K. K. (2018). Perinatal sucrose exposure in

rats disrupts hormones, brain, and behavior in adulthood. Presented at Neuroscience 2018, San Diego, CA. (Credited on poster, not on abstract).

18. **Forys, B.**, Phi, J., Shi, L., Yu, V. ZH. (2018). Emojinal perception: Emoji presence and perceived emotional valence. Presented at UBC Psychology Undergraduate Research Conference, 2018 (PSYC 217 Poster Award, 1st place winner).
19. **Forys, B.**, Tandun, R., Cookson, J., & Xiao, D. (2018). Predicting facial and paw movement from cortical mesoscopic calcium activity in mice: A machine learning perspective. Presented at UBC Multidisciplinary Undergraduate Research Conference, 2018.

**Trainee

INVITED TALKS

1. **Forys, B. J.** (2025). From optimality to experience: Cognitive control, modelling neural dynamics, and beyond. Delivered to Poldrack Lab, Stanford University, Stanford, CA (virtual).
2. **Forys, B. J.** (2025). From fMRIprep to tedana, and validating multi-echo task data using CPCA. Delivered to Tedana multi-echo fMRI working group (virtual).
3. **Forys, B. J.** (2023). Exploring the structure of brain activity (and why it matters for neurophenomenology). Delivered at UBC Neurophenomenology Meeting, 2023, Vancouver, BC.
4. **Forys, B. J.** & Todd, R. M. (2023). Using fMRI to model representational content in a translational context. Delivered at UBC MRI Research Centre Annual Retreat, 2023, Vancouver, BC.

CONFERENCE TALKS

1. **Forys, B. J.**, Moise, N., Todd, R. M., & Winstanley, C. A. (2026). Weighting the work: Similar factors predict cognitive effort preferences when working for rewards vs. Avoiding losses. Delivered at Northwest Cognition and Memory (NOWCAM), 2026, Vancouver, BC.
2. **Forys, B. J.**, Kryklywy, J. H., Ehlers, M. R., Moore, S. R., Beukers, A. O., Anderson, A. K., & Todd, R. M. (2022). Modelling representations of continuously shifting stimuli: a novel sliding window approach. Delivered at Cognitive Neuroscience Society Annual Meeting, 2022, San Francisco, CA.

SOFTWARE

1. Xiao, D., **Forys, B. J.**, Vanni, M. P. & Murphy, T. H. (2021). MesoNet: automated scaling and segmentation of mouse mesoscale cortical maps using machine learning. Code Ocean. <https://doi.org/10.24433/CO.1919930.V1>
2. Kryklywy, J.H., **Forys, B. J.**, & Todd, R.M. (2021). Pattern Component Modelling for R (PCM-R), R package, pre-release. <https://github.com/bf777/PCMforR>

DATASETS

1. Ehlers, M. R., Kryklywy, J. H., Beukers, A. O., Moore, S. R., **Forys, B. J.**, Anderson, A.K. & Todd, R. M. (2024). Valenced tactile information is evoked by neutral visual cues following emotional learning. OpenNeuro. <https://doi.org/10.18112/openneuro.ds005449.v1.0.0>

Teaching, mentoring, and organization

TEACHING

Sessional Lecturer, PSYC 365, Cognitive Neuroscience, UBC 2025
Taught, developed the course material and assessments for, and supervised TAs for the main undergraduate cognitive neuroscience course at UBC Vancouver, a class of 220 students.

Teaching Fellow, Dr. Jill Dosso, PSYC 217, Research Methods in Psych., UBC 2025
Senior Teaching Fellow, Dr. Catherine Rawn, PSYC 217, Research Methods in Psych., UBC 2024

Teaching Assistant and Guest Lecturer, UBC fMRI Brain Camp 2023-2024
Co-designed and conceptualized the course structure. Taught five sessions on fMRI preprocessing and open data practices, including on how to use the BIDS format, conduct quality control, run representational similarity analysis, use fMRIPrep, and run analyses on high-performance computing platforms. Provided technical support to students at all stages of preprocessing.

Teaching Fellow, Dr. Andrew Rivers, PSYC 217, Research Methods in Psych., UBC 2022
Teaching Assistant, Dr. Rebecca Todd, PSYC 365, Cognitive Neuroscience, UBC 2021-2024, 2026
Teaching Fellow, Dr. Catherine Rawn, PSYC 217, Research Methods in Psych., UBC 2020-2021, 2023

Teaching Fellow, Dr. Catherine Rawn, PSYC 218, Analysis of Behavioural Data, UBC 2020
Teaching Assistant, Dr. David King, PSYC 305A, Personality Psych., UBC 2019

Guest Lecturer, PSYC 365, Dr. Rebecca Todd, Cognitive Neuroscience, UBC 2026
Guest Lecturer, PSYC 571, Dr. Rebecca Todd, Beyond the individual brain: Interactive social cognition in an enactive framework, UBC 2025
Guest Lecturer, PSYC 539, Dr. Noah Silverberg, Biological bases of mental disorders, UBC 2025
Guest Lecturer, PSYC 333, Dr. Daniela Palombo, Memory 1, UBC 2023-2025

MENTORING

Undergraduate honours students

Michelle Gitaari, University of British Columbia 2023-2024
Honours thesis: *High vs. low effort: How perception of cognitive effort ability impacts effort deployment choices.*

Skye Zheng, University of British Columbia 2022-2023
Honours thesis: *Maladaptive transfer of cognitive effort in emotional situations.*

Undergraduate directed studies students

Dayana Stamboliyska, University of British Columbia 2021-2022
Paper: *Self-Efficacy in modern computational theories of motivated action.*

Undergraduate research assistants

Nick Moise, University of British Columbia 2025-present
Anisha Singh, University of British Columbia 2024-present
Emily Qi, University of British Columbia 2023-2024
Rita Jin, University of British Columbia 2023-2024
Niharika Dwivedi, University of British Columbia 2023
Paul Kee, University of British Columbia 2022
Aanandi Sidarth, University of British Columbia 2021-2022

Mentorship positions

Diversity Mentor, UBC Psychology Diversity Mentorship Program 2021-present
Mentors students interested in applying to graduate school in Psychology, guiding them through the mentor selection and school application process and meeting with them on a regular basis.

Reviewer activities

Reviewed articles for the following journals:

eLife
Nature
Journal of Environmental Psychology
PLOS Mental Health
Imaging Neuroscience

ORGANIZATION AND LEADERSHIP

Organizing Committee Member, 5th International Sleep Replay Workshop 2026
Conducted registration, wayfinding, and support at the 5th International Sleep Replay Workshop hosted by UC Irvine at the Cognitive Neuroscience Society annual meeting in 2026.
PsychFest Committee Member, UBC Psychology 2024-2025
Selected and booked the venue for, and helped organize, the PsychFest end-of-year conference for UBC Psychology graduate students. Served on the UBC Psychology Graduate Student Council as part of this role.

Skills

Programming Languages: Python, bash, R, JavaScript, MATLAB, C#, HTML/CSS, Swift, Java
Machine Learning and Computer Vision: TensorFlow, Keras, OpenCV
Experimentation Technology: PsychoPy, PsychToolbox, fMRIPrep, AFNI, FSL, CONN, fMRI-CPCA, BioSemi EEG, EEGLAB, LabChart/PowerLab

Web Frameworks: Django, Flask, npm, Bootstrap, Wordpress, Squarespace
Infrastructure: Docker, Azure, High performance computing (SLURM, PBS)
Visualization: Photoshop, Illustrator, Inkscape, Unity, Blender, AutoCAD, 3DSMax, Revit, Inventor

Professional affiliations

Member, Psychonomic Society	2025-present
Member, Society for Neuroscience	2019-present
Member, Psi Chi International Honor Society in Psychology	2019-present

Languages

English – native; French – fluent